A Consolidated Adelic Spectral-Flow Program for the Riemann Hypothesis

Li Positivity, Sonine Compression, b-Calculus, Witt Gauge Geometry, the Renormalized Boundary Defect, and the Gamma-Compressed Kernel Obstruction

Synthesized from the supplied manuscripts and subsequent analysis

Attribution: original program authors and collaborative AI-review extensions;
this extension adds the gamma-compressed Weil-kernel positivity analysis and negative-support obstruction.

April 2026 synthesis; extended version

Abstract

This paper gives a self-contained synthesis of a sequence of supplied manuscripts developing a proposed program toward the Riemann Hypothesis. The program attempts to unify Connes' noncommutative geometry of the adele-class space, the Berry–Keating $H = xp$ picture, $\mathbb{F}_1$-geometric positivity, Sonine-space compression, and supergeometric spectral flow through Atiyah–Patodi–Singer (APS) boundary theory. The present article does not claim a proof of the Riemann Hypothesis. Instead it isolates the strongest coherent theorem stack that emerges from the program and makes explicit where the remaining RH-level difficulty resides.

The main consolidation is this: Li positivity may be attacked through an APS/spectral-flow formula of the schematic form

$$\lambda_n = I_n - \frac{1}{2}h_n,$$

where $h_n = \dim\ker D_n$ is the zero-mode obstruction and I_n is a boundary-renormalized spectral invariant. The zero-mode problem is reduced to a b-Weitzenboeck coercivity inequality. The positivity of the boundary invariant is reformulated through a gamma-weighted Sonine compression and a Schur/Herglotz boundary transfer function. A b-regularized compactness theorem is proposed for the gamma-Sonine boundary projection. A crucial correction is also made: the ordinary Fredholm pair index is integer-valued and therefore cannot literally equal the real Archimedean term in the Weil formula. The appropriate object is a renormalized relative eta or spectral-shift defect.

This extended version adds a further obstruction theorem. The gamma-compressed Weil kernel is positive definite only if the gamma-Sonine projection removes all negative spectral directions of the Mellin multiplier, not merely the scalar Archimedean trace defect. Finite gamma-moment compression cannot generally achieve this when the negative set has positive measure. Thus the next exact target is a strong spectral-support theorem, $P_\Gamma \le 1_{\{m_n \ge 0\}}(M_{m_n})$, or an equivalent domination inequality on the gamma-Sonine subspace.

Contents

1 Attribution, scope, and status **3**

1.1 The central conclusion 3

2 Classical input: Li coefficients and the Weil decomposition **4**

2.1 Li's criterion 4

2.2 The Weil decomposition 4

3 The unified geometric architecture **4**
3.1 Resolution I: Sonine compression . . . 5
3.2 Resolution II: graded Clifford-adeles and spectral flow . . . 5

4 What the five manuscripts contributed **5**
4.1 Original Vision Paper . . . 5
4.2 Revision 1 / Version 3 . . . 6
4.3 Revision 2 / Version 5 . . . 6
4.4 Quantum proposals, Versions 6.3 and 6.4 . . . 6

5 The minimal theorem stack **6**

6 The zero-mode problem via b-Weitzenboeck coercivity **7**
6.1 Li–Witt convexity and the Riccati route . . . 8

7 Boundary positivity and the Schur/Herglotz route **9**
7.1 The Li generating series and disk transfer functions . . . 9
7.2 Why ordinary Sonine compression is insufficient . . . 9
7.3 Gamma-weighted Sonine compression . . . 9

8 Compactness of the b-regularized gamma-Sonine projection **10**
8.1 Boundary model . . . 10
8.2 Renormalized gamma defect . . . 10

9 The index mismatch and the renormalized eta correction **11**
9.1 Renormalized relative eta defect . . . 11

10 Revised statement of Lemma B **12**

11 The gamma-compressed Weil kernel and the new obstruction **12**
11.1 Mellin multiplier form . . . 13
11.2 Finite gamma compression cannot generally prove positivity . . . 13
11.3 Negative-support exclusion is insufficient by itself . . . 14
11.4 The exact revised target . . . 14

12 Consequences for the zero-mode route **14**

13 The quantum branch as a dependent application **15**

14 Falsifiable research program **15**
14.1 Finite-dimensional truth laboratory . . . 15
14.2 Analytic tasks . . . 16

15 Final consolidated theorem **16**

16 Conclusion **16**

1 Attribution, scope, and status

This paper consolidates and extends five supplied manuscripts:

(i) *A Vision Paper: A Grand Unified Program for the Riemann Hypothesis via $\mathbb{F}_1$ Geometry, Berry–Keating Scaling, and Supergeometric Spectral Flow*, here cited as the Original Vision Paper.

(ii) *A Grand Unified Program for the Riemann Hypothesis, Version 3: Advances on All Ten Research Tasks*, here cited as Revision 1 or Version 3.

(iii) *A Grand Unified Program for the Riemann Hypothesis, Version 5: b-Calculus and Weitzenboeck Geometry*, here cited as Revision 2 or Version 5.

(iv) *A Grand Unified Program for the Riemann Hypothesis, Version 6.3: Effective Adelic Spectral Flow, Quantum Register Embedding, and an Intrinsically Fault-Tolerant Topological Factoring Oracle*, here cited as Proposal Foundations.

(v) *A Grand Unified Program for the Riemann Hypothesis, Version 6.4: Effective Adelic Spectral Flow, Quantum Register Embedding, and an Intrinsically Fault-Tolerant Topological Factoring Oracle*, here cited as Quantum Proposal.

The present synthesis also incorporates subsequent analysis: the "dream theorem" reduction to b-Weitzenboeck coercivity, the gamma-weighted Sonine projection, the Schur/Herglotz boundary reformulation, the compactness route for the b-regularized gamma correction, the correction distinguishing an integer Fredholm index from a real renormalized eta defect, and the newly isolated obstruction to positive definiteness of the gamma-compressed Weil kernel.

Remark 1.1 (No proof claim). No proof of the Riemann Hypothesis is claimed. Several statements below are conditional. The paper's purpose is to form one cohesive and rigorous research program, state the best available conditional theorem stack, and isolate the exact points where a proof would still need to be supplied.

1.1 The central conclusion

The best consolidated program is not the quantum-factoring branch. The strongest mathematical core is

$$\text{Li positivity} \quad \Longleftarrow \quad \text{APS boundary positivity + zero-mode vanishing.}$$

More explicitly, the program seeks a formula of the form

$$\lambda_n = I_n - \frac{1}{2}h_n, \qquad h_n = \dim\ker(D_n), \tag{1}$$

where λ_n are the Li coefficients, $D_n = D_0 + A(f_n)$ is a gauge-deformed Dirac-type operator, and I_n is the relevant APS/spectral-flow invariant. If

$$h_n = 0, \qquad I_n \geq 0$$

for every $n \geq 1$, then Li's criterion gives RH.

The essential work splits into two tasks:

Task 1. Prove the zero-mode theorem $\ker D_n = 0$.

Task 2. Prove positivity of the correct boundary invariant I_n.

The rest of this paper develops the most precise known formulation of these two tasks.

2 Classical input: Li coefficients and the Weil decomposition

2.1 Li's criterion

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

be the completed zeta function. Li's coefficients are

$$\lambda_n = \sum_{\rho}\left[1 - \left(1 - \frac{1}{\rho}\right)^n\right], \tag{2}$$

where the sum ranges over non-trivial zeros of $\zeta(s)$ in the usual symmetric sense. Equivalently,

$$\lambda_n = \frac{1}{(n-1)!}\frac{d^n}{ds^n}\left[s^{n-1}\log\xi(s)\right]_{s=1}. \tag{3}$$

Li's theorem states

$$RH \quad \Longleftrightarrow \quad \lambda_n \geq 0 \quad \text{for all } n \geq 1. \tag{4}$$

This is the arithmetic target of the whole program.

2.2 The Weil decomposition

The Original Vision Paper emphasizes that the explicit formula gives, for normalized Li probes f_n,

$$\lambda_n = A_n + B_n + C_n. \tag{5}$$

In the normalized case $C_n = 0$, so

$$\lambda_n = A_n + B_n. \tag{6}$$

Here B_n is the positive prime-orbit contribution and A_n is the negative Archimedean contribution. In schematic form,

$$B_n = \sum_p \sum_{m\geq 1} \frac{\log p}{p^{m/2}}\left(f_n(p^m) + f_n(p^{-m})\right) > 0, \tag{7}$$

while

$$A_n = -\frac{1}{2\pi}\int_{-\infty}^{\infty} f_n\left(\frac{1}{2}+it\right)\frac{\Gamma'}{\Gamma}\left(\frac{1}{4}+\frac{it}{2}\right)dt < 0. \tag{8}$$

Thus RH is not equivalent to mere positivity of the prime side. It requires the domination

$$B_n \geq |A_n| \qquad \forall n \geq 1. \tag{9}$$

This is the basic obstruction around which every subsequent construction must be organized.

3 The unified geometric architecture

The Original Vision Paper identifies three major lines of thought and proposes that they are different faces of the same structure:

1. Connes' noncommutative geometry on the adele-class space $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$.
2. The Berry–Keating quantization of the Hamiltonian $H = xp$.
3. $\mathbb{F}_1$-geometric intersection positivity.

The proposed bridge is the multiplicative scaling action on the adele-class space. If $u \in \mathbb{R}_+^\times$, the scaling operator on a suitable Hilbert space $\mathcal{H} = L^2(X, d^*x)$ is

$$U(f)\psi(x) = \int_{\mathbb{R}_+^\times} f(u)\psi(xu)\frac{du}{u}. \tag{10}$$

Its infinitesimal generator is the Berry–Keating operator

$$D = x\frac{d}{dx} + \frac{1}{2}. \tag{11}$$

The regularized trace of $U(f_n)$ recovers the Weil distribution:

$$\mathrm{Tr}_{\mathrm{reg}}\, U(f_n) = \lambda_n. \tag{12}$$

But because $A_n < 0$, $U(f_n)$ cannot be globally positive on the full Hilbert space. The program therefore proposes two complementary repairs.

3.1 Resolution I: Sonine compression

Let P_{Son} be a projection onto a Sonine-type subspace of Mellin test functions. The original conjecture is that the compressed operator is positive:

$$P_{\mathrm{Son}} U(f_n) P_{\mathrm{Son}} \geq 0. \tag{13}$$

This is meant to remove or control the negative Archimedean scattering contribution. It is a falsifiable operator inequality: finite truncations should reveal any persistent negative eigenvalues.

3.2 Resolution II: graded Clifford-adeles and spectral flow

The second mechanism introduces a $\mathbb{Z}_2$-graded Hilbert module

$$\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-, \tag{14}$$

where the positive component models prime orbits and the negative component models Archimedean scattering. The ordinary trace is replaced by a supertrace:

$$\mathrm{STr}(U(f_n)) = \mathrm{Tr}_{\mathcal{H}_+}(U(f_n)) - \mathrm{Tr}_{\mathcal{H}_-}(U(f_n)) = B_n - |A_n| = \lambda_n. \tag{15}$$

A Clifford-valued Dirac operator D_0 is then deformed by a Witt-vector gauge potential:

$$D_n = D_0 + A(f_n). \tag{16}$$

RH is recast as a positivity or monotonicity statement for the spectral flow of the family $D_t = D_0 + tA(f_n)$.

4 What the five manuscripts contributed

4.1 Original Vision Paper

The Original Vision Paper is conceptually strongest where it is cautious. It states no proof claim, identifies the Archimedean term as the obstruction, and proposes two falsifiable paths: Sonine compression and supergeometric spectral flow. It also identifies a McKean–Singer obstruction to a static factorization and therefore motivates a dynamic spectral-flow formulation.

4.2 Revision 1 / Version 3

Revision 1 adds three important components:

1. a $\mathbb{Z}_2$-grading regularity statement;
2. the vanishing of the eta invariant $\eta(D_0) = 0$ by spectral symmetry;
3. an APS formula of the schematic form

$$\lambda_n = \mathrm{ind}_{\mathrm{APS}}(D_0, P_n) - \frac{1}{2}h_n, \qquad h_n = \dim\ker(D_n). \tag{17}$$

This identifies $h_n = 0$ as the decisive obstruction. Revision 1 also reports positivity of computed Li coefficients up to $n = 100$, but those numerics are sanity checks rather than evidence at the level needed for RH.

4.3 Revision 2 / Version 5

Revision 2 is the strongest analytic revision. It introduces b-calculus and weighted Sobolev spaces to handle the noncompact adele-class space. The Archimedean component is treated as a cylindrical end, and the Fredholm property is restored by b-ellipticity and an invertible indicial family. It also develops a Weitzenboeck approach to $h_n = 0$ and states a generating-series reformulation:

$$G(z) = \sum_{n\geq 1} \lambda_n z^n. \tag{18}$$

The proposed singularities are

$$z_\rho = \frac{\rho}{\rho - 1}, \tag{19}$$

with $|z_\rho| = 1$ if and only if $\mathrm{Re}(\rho) = 1/2$. Thus RH is equivalent to analyticity of G in $|z| < 1$, subject to the necessary convergence and analytic-continuation controls.

4.4 Quantum proposals, Versions 6.3 and 6.4

The quantum branch proposes an $O(\log N)$-qubit embedding of a localized Bost–Connes module, adiabatic deformation by a Witt gauge potential, quantum phase estimation of an APS index, and a topologically protected factoring oracle. Its conceptual contribution is the idea that a topological invariant may be robust to local noise.

However, this branch remains speculative for two reasons. First, the polynomial-time claim depends on a conjectural adiabatic spectral gap. Second, the readout appears to require efficient localization on prime-factor support without already knowing the factors. Therefore, the quantum material should be treated as a possible application of a future analytic theorem, not as the foundation of the RH program.

5 The minimal theorem stack

The consolidated program should now be stated as a theorem stack.

Conjecture 5.1 (APS-Li formula)**.** *For each Li probe f_n there is a b-Fredholm Dirac-type operator D_0, an admissible APS boundary projection P_n, and a gauge-deformed operator $D_n = D_0 + A(f_n)$ such that*

$$\lambda_n = I_n - \frac{1}{2}h_n, \qquad h_n = \dim\ker(D_n), \tag{20}$$

where I_n is the appropriate renormalized APS/spectral-flow invariant.

Conjecture 5.2 (Zero-mode theorem)**.** *For every $n \geq 1$,*

$$\ker(D_n) = \{0\}. \tag{21}$$

Conjecture 5.3 (Boundary positivity)**.** *For every $n \geq 1$,*

$$I_n \geq 0. \tag{22}$$

Theorem 5.4 (Conditional RH theorem)**.** *Assume Conjectures 5.1, 5.2, and 5.3. Then RH holds.*

Proof. By Conjecture 5.2, $h_n = 0$ for all n. By Conjecture 5.1, $\lambda_n = I_n$. By Conjecture 5.3, $\lambda_n \geq 0$ for all n. Li's criterion gives RH. □

6 The zero-mode problem via b-Weitzenboeck coercivity

The zero-mode obstruction is the most concrete part of the program. On the p-sector of the Bost–Connes–Clifford module and in the b-cylinder coordinate $u = \log x$, take the local normal form

$$D_{n,p} = \gamma^1 \log p + \gamma^2 \left(\partial_u + \frac{1}{2} + a_{n,p}(u)\right), \tag{23}$$

with Clifford relations

$$(\gamma^1)^2 = (\gamma^2)^2 = 1, \qquad \gamma^1\gamma^2 + \gamma^2\gamma^1 = 0. \tag{24}$$

The scalar $a_{n,p}(u)$ is a local representative of the Witt gauge potential. A natural regularized representative is

$$a_{n,p}(u) = (\log p)\chi(u) \operatorname{Re} f_n(pe^u), \tag{25}$$

where $\chi \in C_c^\infty(\mathbb{R})$ is a b-cylinder cutoff.

Proposition 6.1 (Local b-Weitzenboeck formula)**.** *For $D_{n,p}$ as in (23),*

$$D_{n,p}^* D_{n,p} = -\partial_u^2 + (\log p)^2 + \left(\frac{1}{2} + a_{n,p}(u)\right)^2 - a_{n,p}'(u). \tag{26}$$

Proof. The formal adjoint in the b-measure is

$$D_{n,p}^* = \gamma^1 \log p + \gamma^2 \left(-\partial_u + \frac{1}{2} + a_{n,p}(u)\right).$$

The mixed Clifford terms cancel by anticommutation. Writing $b(u) = 1/2 + a_{n,p}(u)$,

$$(-\partial_u + b)(\partial_u + b) = -\partial_u^2 + b^2 - b'.$$

This gives (26). □

Theorem 6.2 (Conditional zero-mode theorem)**.** *Suppose that, for every n, there exists $\kappa_n > 0$ such that*

$$(\log p)^2 + \left(\frac{1}{2} + a_{n,p}(u)\right)^2 - a_{n,p}'(u) \geq \kappa_n \tag{27}$$

for all relevant primes p and all u in the b-cylinder model. Then $\ker(D_n) = \{0\}$.

Proof. The global b-Weitzenboeck formula has the form

$$D_n^* D_n = (\nabla^{(n)})^* \nabla^{(n)} + V_n,$$

with V_n locally given by the scalar potential in (27). The APS boundary contribution is nonnegative because the APS projection removes the negative spectral modes of the boundary operator. Therefore, for ψ in the APS/b-domain,

$$\|D_n\psi\|^2 \geq \|\nabla^{(n)}\psi\|^2 + \kappa_n\|\psi\|^2 \geq \kappa_n\|\psi\|^2.$$

If $D_n\psi = 0$, then $\psi = 0$. □

Since $p \geq 2$, a sufficient form of (27) is

$$\sup_{p,u}\left[a'_{n,p}(u) - \left(\frac{1}{2} + a_{n,p}(u)\right)^2\right] < (\log 2)^2. \tag{28}$$

This is the Li–Witt coercivity inequality. It is the most concrete formulation of the zero-mode problem currently available.

6.1 Li–Witt convexity and the Riccati route

Let

$$b_{n,p}(u) = \frac{1}{2} + a_{n,p}(u).$$

The sharper Riccati inequality

$$b'_{n,p}(u) \leq b_{n,p}(u)^2 \tag{29}$$

implies (28), since then the potential in (26) is at least $(\log p)^2 \geq (\log 2)^2$.

A useful sufficient mechanism is an outer-amplitude representation. Suppose

$$b_{n,p}(u) = -\frac{d}{du}\log y_{n,p}(u), \qquad y_{n,p}(u) > 0. \tag{30}$$

Then

$$b'_{n,p}(u) = b_{n,p}(u)^2 - \frac{y''_{n,p}(u)}{y_{n,p}(u)}. \tag{31}$$

Thus $y''_{n,p}(u) \geq 0$ implies the Riccati inequality.

Proposition 6.3 (Positive spectral measure implies Li–Witt convexity)**.** *If the gamma-compressed prime-sector amplitude has the Laplace representation*

$$y_{n,p}(u) = \int_0^\infty e^{-u\xi}\, d\mu_{n,p}(\xi), \qquad \mu_{n,p} \geq 0, \tag{32}$$

then $y''_{n,p}(u) \geq 0$. *Consequently* (29) *holds and* $D_n^* D_n \geq (\log 2)^2 I$.

Proof. Differentiating under the integral gives

$$y''_{n,p}(u) = \int_0^\infty \xi^2 e^{-u\xi}\, d\mu_{n,p}(\xi) \geq 0.$$

Then (30) gives (29); Proposition 6.1 gives the lower bound. □

Thus the zero-mode problem is converted into a positive-measure problem for the gamma-compressed prime-sector amplitude. The subsequent sections analyze whether the gamma compression really supplies such a positive measure.

7 Boundary positivity and the Schur/Herglotz route

The boundary positivity conjecture should not be phrased as positivity of an ordinary integer APS index. The Li coefficients and Archimedean terms are real-valued. Therefore, the correct object must be a renormalized APS invariant: a relative eta, a spectral-shift functional, or a renormalized supertrace.

7.1 The Li generating series and disk transfer functions

Revision 2 introduced the generating function

$$G(z) = \sum_{n\geq 1} \lambda_n z^n.$$

A more structural disk object is

$$F(z) = \xi\left(\frac{1}{1-z}\right). \tag{33}$$

Then

$$\frac{d}{dz}\log F(z) = \sum_{n\geq 1} \lambda_n z^{n-1}, \qquad z\frac{F'(z)}{F(z)} = G(z). \tag{34}$$

Thus the Li generating series is a logarithmic derivative of a disk-transformed completed zeta function.

The Schur/Herglotz route asks whether the associated boundary transfer function is contractive in the unit disk. A generic Schur function Θ is characterized by positivity of the de Branges–Rovnyak kernel

$$K_\Theta(z,w) = \frac{1-\Theta(z)\overline{\Theta(w)}}{1-z\overline{w}}. \tag{35}$$

If the Li boundary transfer function is Schur, the corresponding Toeplitz/relative boundary object has a positive orientation. This is the analytic form of APS boundary positivity.

7.2 Why ordinary Sonine compression is insufficient

The original Sonine condition annihilates Mellin data near the critical line, but the Archimedean contribution (8) involves the entire gamma kernel

$$\frac{\Gamma'}{\Gamma}\left(\frac{1}{4}+\frac{it}{2}\right).$$

Therefore, ordinary Sonine compression does not obviously remove the full negative Archimedean term.

7.3 Gamma-weighted Sonine compression

Define the Archimedean gamma functional

$$\mathcal{A}_\Gamma(\varphi) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \varphi(t)\frac{\Gamma'}{\Gamma}\left(\frac{1}{4}+\frac{it}{2}\right)dt. \tag{36}$$

The gamma-weighted Sonine space is the closed subspace

$$\mathcal{S}_\Gamma = \{\varphi : \mathcal{A}_\Gamma(\varphi) = 0\}, \tag{37}$$

or, in a finite-mode model, the common kernel of finitely many regularized gamma moment functionals. Let P_Γ be the projection onto this space. The intended effect is

$$P_\Gamma(A\text{-term})P_\Gamma = 0, \qquad P_\Gamma U(f_n)P_\Gamma \sim \text{positive prime contribution.} \tag{38}$$

This is the correct target for Lemma B: it attacks exactly the negative term rather than a vague neighborhood of the critical line.

8 Compactness of the b-regularized gamma-Sonine projection

We now state the strongest rigorous bridge between the Li APS boundary condition P_n and the gamma-weighted Sonine projection P_Γ.

8.1 Boundary model

Let

$$\mathcal{H}_\partial = L^2(\mathbb{R}_t; \mathcal{K}) \tag{39}$$

be the Mellin-side boundary Hilbert space, where t is the Fourier-Mellin variable of the Archimedean cylinder and $\mathcal{K}$ is the prime-Clifford multiplicity space. Let B_n be the Li boundary operator and

$$P_n = 1_{[0,\infty)}(B_n). \tag{40}$$

The raw gamma multiplier

$$M_\psi(t) = \frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{it}{2}\right) \tag{41}$$

is not compact. Indeed, as $|t| \to \infty$,

$$\frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{it}{2}\right) = \log|t| + O(|t|^{-1}). \tag{42}$$

Multiplication by $\log|t|$ on $L^2(\mathbb{R})$ is not compact. Therefore the full gamma multiplier cannot be used as a compact perturbation.

8.2 Renormalized gamma defect

Separate the principal asymptotic part and retain only the short-range defect:

$$m_\Gamma(t) = \frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{it}{2}\right) - \log(1 + |t|). \tag{43}$$

Then $m_\Gamma(t) = O(|t|^{-1})$ at infinity, modulo lower-order regularization near $t = 0$. Let $\chi \in C_c^\infty(\mathbb{R})$ be a spectral cutoff. Define

$$R_\Gamma = \chi(B_n) M_{m_\Gamma} \chi(B_n), \tag{44}$$

and

$$P_\Gamma = 1_{[0,\infty)}(B_n + R_\Gamma). \tag{45}$$

Theorem 8.1 (Compactness of the b-regularized gamma correction)**.** *In the b-regularized boundary model above,*

$$R_\Gamma \in \mathcal{K}(\mathcal{H}_\partial). \tag{46}$$

Proof. The operator $\chi(B_n)$ is spectrally compactly supported and smoothing in the b-calculus. In the exact cylinder model, it is a Fourier multiplier with smooth compact support; in the geometric b-model it has a smooth b-kernel with compactly supported normal operator. The multiplier M_{m_Γ} is bounded on the support selected by χ. Hence $\chi(B_n)M_{m_\Gamma}\chi(B_n)$ is Hilbert–Schmidt in the cylinder model and a compact residual b-operator in the geometric model. Thus R_Γ is compact. □

Theorem 8.2 (Compactness of the projection difference)**.** *Assume the b-weight is chosen so that B_n has an essential spectral gap around zero and the path $B_t = B_n + tR_\Gamma$ has no essential-spectrum crossing at zero. Then*

$$P_\Gamma - P_n \in \mathcal{K}(\mathcal{H}_\partial). \tag{47}$$

Proof. For $z \notin \mathbb{R}$, the resolvent identity gives

$$(B_n + R_\Gamma - z)^{-1} - (B_n - z)^{-1} = -(B_n + R_\Gamma - z)^{-1} R_\Gamma (B_n - z)^{-1}.$$

Since R_Γ is compact and the resolvents are bounded, the resolvent difference is compact. Choose a smooth function f equal to 0 below $-\varepsilon$ and 1 above ε, where $(-\varepsilon, \varepsilon)$ misses the essential spectrum. By the Helffer–Sjoestrand functional calculus, $f(B_n + R_\Gamma) - f(B_n)$ is compact. The difference between $f(B)$ and the sharp projection $1_{[0,\infty)}(B)$ occurs only in the finite-dimensional discrete spectral window. Therefore $P_\Gamma - P_n$ is compact. □

Corollary 8.3 (Fredholm pair)**.** *The pair (P_Γ, P_n) is a Fredholm pair. Consequently P_n and P_Γ lie in the same restricted Grassmannian up to finite spectral flow.*

9 The index mismatch and the renormalized eta correction

The compactness theorem enables a relative Fredholm-pair index

$$\operatorname{ind}(P_\Gamma, P_n) = \dim(\operatorname{Ran} P_\Gamma \cap \ker P_n) - \dim(\ker P_\Gamma \cap \operatorname{Ran} P_n). \tag{48}$$

Equivalently,

$$\operatorname{ind}(P_\Gamma, P_n) = \operatorname{sf}(B_n + tR_\Gamma; 0) \tag{49}$$

up to orientation. This is an integer.

This exposes a serious correction. The Archimedean term A_n is a real number, not generally an integer. For example λ_1 is approximately 0.023096. Therefore an ordinary Fredholm index cannot literally equal λ_n or A_n. Any formula claiming (17) with an ordinary integer-valued index must be reinterpreted. The invariant must be renormalized, real-valued, or spectral-shift-like.

9.1 Renormalized relative eta defect

Define

$$I_{\mathrm{ren}}(P_\Gamma, P_n) = -\frac{1}{2}\left(\eta_{\mathrm{ren}}(B_n + R_\Gamma) - \eta_{\mathrm{ren}}(B_n)\right). \tag{50}$$

The eta variation formula is formally

$$\frac{d}{dt}\eta_{\mathrm{ren}}(B_t) = -\frac{2}{\sqrt{\pi}} \operatorname{Tr}_{\mathrm{ren}}\left(R_\Gamma e^{-B_t^2}\right). \tag{51}$$

Integrating from 0 to 1 and choosing R_Γ to be the b-regularized gamma defect, the renormalized trace recovers the Archimedean gamma term in the Weil formula. Thus the correct target is

$$-\frac{1}{2}\left(\eta_{\mathrm{ren}}(B_n + R_\Gamma) - \eta_{\mathrm{ren}}(B_n)\right) = -A_n = |A_n|. \tag{52}$$

This is not an ordinary index theorem. It is a spectral-shift/eta-defect theorem.

Proposition 9.1 (Corrected Archimedean subtraction principle)**.** *Assume the renormalized eta variation formula is valid for the b-regularized gamma path and that R_Γ is normalized to represent the short-range gamma defect in the Weil formula. Then*

$$\lambda_n + I_{\mathrm{ren}}(P_\Gamma, P_n) = B_n > 0. \tag{53}$$

Equivalently,

$$\lambda_n = B_n - I_{\mathrm{ren}}(P_\Gamma, P_n). \tag{54}$$

Proof. Using $\lambda_n = A_n + B_n$ and (52),

$$\lambda_n + I_{\mathrm{ren}} = (A_n + B_n) + (-A_n) = B_n.$$

The prime contribution B_n is positive in the prime-orbit representation. □

Remark 9.2 (What this does and does not prove)*.* The gamma-Sonine correction removes the negative Archimedean contribution and leaves the positive prime contribution. This does not prove RH by itself, because RH requires $B_n \geq |A_n|$ before removal. The computation clarifies the mechanism but does not replace the domination inequality.

10 Revised statement of Lemma B

The earlier Lemma B asked for APS boundary positivity. The corrected form is:

Conjecture 10.1 (Renormalized boundary positivity)**.** *For each n, the Li boundary invariant I_n is a renormalized APS/spectral-shift functional satisfying*

$$I_n \geq 0, \tag{55}$$

and it is related to the gamma-Sonine boundary condition by the eta-defect formula

$$I_n + I_{\mathrm{ren}}(P_\Gamma, P_n) = B_n. \tag{56}$$

In this language, RH becomes the assertion that the original Li boundary invariant is nonnegative, while the gamma-Sonine projection gives a positive compressed model after the Archimedean defect has been removed.

11 The gamma-compressed Weil kernel and the new obstruction

This section extends the prior synthesis. It analyzes the operator inequality

$$P_\Gamma U(f_n) P_\Gamma \geq 0, \tag{57}$$

which was previously proposed as the key input for the positive-measure route to Li–Witt convexity and then to the zero-mode theorem.

11.1 Mellin multiplier form

On $\mathbb{R}_+^\times$, the Mellin transform diagonalizes multiplicative convolution. Let

$$(\mathcal{M}\psi)(t) = \int_0^\infty \psi(x) x^{-1/2-it} \frac{dx}{x}. \tag{58}$$

Then

$$\mathcal{M}U(f_n)\mathcal{M}^{-1} = M_{m_n}, \tag{59}$$

where

$$m_n(t) = \int_0^\infty f_n(u) u^{-1/2-it} \frac{du}{u} \tag{60}$$

up to the normalization chosen for the Li probe. Thus (57) is exactly

$$\int_{\mathbb{R}} m_n(t)|\phi(t)|^2 dt \geq 0 \qquad \forall \phi \in \operatorname{Ran} P_\Gamma. \tag{61}$$

Proposition 11.1 (Exact positivity criterion)**.** *Let P_Γ be an orthogonal projection on the Mellin boundary Hilbert space. Then*

$$P_\Gamma U(f_n) P_\Gamma \geq 0$$

if and only if (61) *holds for every $\phi \in \operatorname{Ran} P_\Gamma$. A sufficient sharp support condition is*

$$P_\Gamma \leq 1_{\{m_n \geq 0\}}(M_{m_n}). \tag{62}$$

Proof. For $\phi \in \operatorname{Ran} P_\Gamma$,

$$\langle P_\Gamma U(f_n) P_\Gamma \phi, \phi \rangle = \langle U(f_n)\phi, \phi \rangle = \int_{\mathbb{R}} m_n(t)|\phi(t)|^2 dt.$$

This proves the equivalence. If (62) holds, then ϕ is supported where $m_n \geq 0$, and the integral is nonnegative. □

11.2 Finite gamma compression cannot generally prove positivity

Let

$$E_- := \{t \in \mathbb{R} : m_n(t) < 0\}. \tag{63}$$

Suppose P_Γ removes only finitely many gamma moments:

$$\operatorname{Ran} P_\Gamma = \{\phi \in L^2(\mathbb{R}) : \langle \phi, g_j \rangle = 0, j = 1, \ldots, r\}. \tag{64}$$

Theorem 11.2 (Finite-codimension obstruction)**.** *If $|E_-| > 0$ and P_Γ is of the finite-codimension form* (64)*, then*

$$\operatorname{Ran} P_\Gamma \cap L^2(E_-) \neq \{0\}. \tag{65}$$

Consequently,

$$P_\Gamma U(f_n) P_\Gamma \ngeq 0. \tag{66}$$

Proof. Since $|E_-| > 0$, $L^2(E_-)$ is infinite-dimensional. Intersecting an infinite-dimensional Hilbert space with the orthogonal complement of finitely many vectors leaves a nonzero vector. Hence there exists $0 \neq \phi \in L^2(E_-)$ with $\langle \phi, g_j \rangle = 0$ for all j. Then $\phi \in \operatorname{Ran} P_\Gamma$ and

$$\langle U(f_n)\phi, \phi \rangle = \int_{E_-} m_n(t)|\phi(t)|^2 dt < 0.$$

Therefore the compressed operator is not positive. □

Corollary 11.3 (Trace subtraction is not operator positivity)**.** *A gamma compression that removes only a scalar Archimedean trace defect A_n does not, by itself, remove all negative spectral directions of M_{m_n}. Therefore it cannot by itself prove positive definiteness of the gamma-compressed Weil kernel.*

11.3 Negative-support exclusion is insufficient by itself

One might try to prove the weaker condition

$$\operatorname{Ran} P_\Gamma \cap L^2(E_-) = \{0\}. \tag{67}$$

This is necessary to rule out vectors supported only in the negative set. But it is not sufficient for positivity. A vector $\phi \in \operatorname{Ran} P_\Gamma$ may have both positive and negative support, $\phi = \phi_+ + \phi_-$, and the negative part may dominate:

$$\int_{E_-} |m_n(t)||\phi(t)|^2 dt > \int_{E_+} m_n(t)|\phi(t)|^2 dt, \tag{68}$$

where $E_+ = \{m_n \geq 0\}$. Thus the true condition is the weighted domination inequality

$$\int_{E_-} |m_n(t)||\phi(t)|^2 dt \leq \int_{E_+} m_n(t)|\phi(t)|^2 dt \quad \forall \phi \in \operatorname{Ran} P_\Gamma. \tag{69}$$

Proposition 11.4 (Equivalent domination form)**.** *The positivity $P_\Gamma U(f_n) P_\Gamma \geq 0$ is equivalent to the domination inequality* (69)*.*

Proof. Split the quadratic form in (61) over E_+ and E_-:

$$\int_{\mathbb{R}} m_n|\phi|^2 = \int_{E_+} m_n|\phi|^2 - \int_{E_-} |m_n||\phi|^2.$$

Nonnegativity for every $\phi \in \operatorname{Ran} P_\Gamma$ is exactly (69). □

11.4 The exact revised target

The gamma-compressed Weil kernel is positive definite if and only if its Mellin multiplier is nonnegative on the gamma-Sonine subspace. The sharp non-circular target is therefore:

$$P_\Gamma \leq 1_{\{m_n \geq 0\}}(M_{m_n}), \tag{70}$$

or, more generally, the domination inequality (69). If P_Γ is defined to be the positive spectral projection of M_{m_n}, then positivity is immediate but circular. The genuine problem is to derive (70) or (69) from the gamma-weighted Sonine/de Branges geometry, without using Li positivity or RH.

12 Consequences for the zero-mode route

The prior zero-mode chain was

$$P_\Gamma U(f_n) P_\Gamma \geq 0 \Rightarrow y''_{n,p} \geq 0 \Rightarrow \text{Riccati coercivity} \Rightarrow D_n^* D_n \geq (\log 2)^2 I \Rightarrow h_n = 0. \tag{71}$$

The new analysis shows that the first implication requires a much stronger operator statement than gamma trace subtraction.

Theorem 12.1 (Corrected zero-mode implication)**.** *Assume the domination inequality* (69) *holds for the gamma-Sonine subspace and that the resulting compressed operator generates a positive spectral measure for the prime-sector amplitude* $y_{n,p}$*. Then* $h_n = 0$*.*

Proof. By Proposition 11.4, $P_\Gamma U(f_n) P_\Gamma \geq 0$. The spectral theorem gives a positive measure representation (32). Proposition 6.3 gives $y''_{n,p} \geq 0$, which gives the Riccati inequality. Proposition 6.1 and Theorem 6.2 then give $h_n = 0$. □

Remark 12.2 (What changed). The gamma-Sonine projection can still be useful, but its role is narrower. It explains how the scalar Archimedean eta defect is subtracted, and it may supply a good candidate boundary condition. It does not automatically supply positivity of the compressed operator. That positivity is equivalent to a full spectral-support or domination theorem.

13 The quantum branch as a dependent application

Versions 6.3 and 6.4 propose a quantum algorithm based on adiabatic spectral flow and quantum phase estimation of an APS index. In the consolidated program, the quantum branch should be reclassified as follows:

1. It is not evidence for RH.
2. It depends on the analytic core: a polynomial spectral gap, a correct gauge-support construction, and a valid topological readout.
3. The APS index robustness idea is conceptually interesting, but it does not remove the need for a rigorous gap theorem and a non-circular construction of the gauge potential.

Therefore the quantum branch should be treated as a possible later application of a solved spectral-flow theory, not as part of the proof of RH.

14 Falsifiable research program

The strongest next steps are concrete.

14.1 Finite-dimensional truth laboratory

Construct a truncation

$$M_{P,L} = \operatorname{span}\{e_p : p \leq P\} \oplus L^2([-L, L]) \otimes \mathbb{C}^2, \tag{72}$$

with explicit boundary conditions and finite matrices for

$$D_0, A(f_n), D_n, P_n, P_\Gamma, R_\Gamma, M_{m_n}.$$

Then test:

1. the smallest eigenvalue of $D_n^* D_n$;
2. the Li–Witt coercivity inequality (28);
3. compactness/norm decay of $P_\Gamma - P_n$ under increasing truncation;
4. numerical validity of the eta-defect identity (52);
5. positivity or failure of the Schur kernel (35);
6. the negative set $E_- = \{m_n < 0\}$ and the dimension/weight of $\operatorname{Ran} P_\Gamma \cap L^2(E_-)$;
7. the domination inequality (69) on the truncated gamma-Sonine subspace.

14.2 Analytic tasks

The paper leaves the following precise open problems.

Problem A: Li–Witt coercivity. Prove (28) for the actual Li probes, or find a counterexample.

Problem B: b-eta variation. Establish the renormalized eta variation formula (51) for the b-regularized gamma path.

Problem C: Schur positivity. Identify the exact Li boundary transfer function and prove positivity of its de Branges–Rovnyak kernel, or show that this is equivalent to RH.

Problem D: domination. Derive $B_n \geq |A_n|$ from the boundary/scattering geometry without assuming Li positivity.

Problem E: index language. Replace all integer-valued APS claims in the earlier manuscripts with the correct real-valued renormalized spectral invariant wherever Li coefficients are being represented.

Problem F: gamma-compressed kernel positivity. Prove either the spectral-support theorem (70) or the domination inequality (69). If finite gamma compression is used, first prove $m_n \geq 0$ a.e.; otherwise Theorem 11.2 gives a negative direction.

15 Final consolidated theorem

Theorem 15.1 (Best current conditional synthesis)**.** *Assume the following six statements.*

(a) The APS-Li formula holds in the corrected renormalized sense:

$$\lambda_n = I_n - \frac{1}{2} h_n.$$

(b) The b-Weitzenboeck coercivity inequality (28) *holds, hence* $h_n = 0$.

(c) The b-regularized gamma-Sonine projection satisfies $P_\Gamma - P_n \in \mathcal{K}(\mathcal{H}_\partial)$.

(d) The renormalized eta-defect identity (52) *holds.*

(e) The original Li boundary invariant satisfies $I_n \geq 0$ *for all* n.

(f) Equivalently or additionally, the gamma-compressed positivity problem is solved by proving the domination inequality (69) *non-circularly on the gamma-Sonine subspace.*

Then RH holds.

Proof. By (b), $h_n = 0$. By (a), $\lambda_n = I_n$. By (e), $\lambda_n \geq 0$ for every n. Li's criterion gives RH. Assumptions (c) and (d) identify how gamma-Sonine compression removes the Archimedean trace defect and supplies the positive compressed model, thereby connecting the Sonine and APS branches. Assumption (f) is the non-trace-level positivity needed for the positive-measure route to zero-mode vanishing. □

16 Conclusion

The five supplied papers contain a promising but overextended program. The best mathematical content is not the quantum factoring proposal; it is the analytic core built around Li coefficients, the Weil decomposition, b-calculus, APS boundary theory, and the zero-mode problem. This synthesis makes five improvements:

1. It reduces the program to a minimal theorem stack.
2. It gives a concrete b-Weitzenboeck coercivity target for $h_n = 0$.
3. It replaces ordinary Sonine compression with gamma-weighted Sonine compression, aimed directly at the negative Archimedean term.
4. It corrects the integer-index mismatch by replacing ordinary APS index language with a renormalized eta/spectral-shift invariant when representing real-valued Li coefficients.
5. It adds the gamma-compressed kernel obstruction: subtracting the Archimedean trace defect is not enough to prove operator positivity. Positive definiteness requires a spectral-support or domination theorem for the Mellin multiplier on the gamma-Sonine subspace.

The most important next theorem is no longer only the b-Weitzenboeck inequality

$$\sup_{p,u}\left[a'_{n,p}(u) - \left(\frac{1}{2} + a_{n,p}(u)\right)^2\right] < (\log 2)^2,$$

combined with the renormalized eta identity

$$-\frac{1}{2}(\eta_{\mathrm{ren}}(B_n + R_\Gamma) - \eta_{\mathrm{ren}}(B_n)) = -A_n.$$

A third theorem is now required:

$$\int_{E_-} |m_n(t)||\phi(t)|^2 dt \le \int_{E_+} m_n(t)|\phi(t)|^2 dt \quad \forall \phi \in \operatorname{Ran} P_\Gamma,$$

or the sharper support condition

$$P_\Gamma \le 1_{\{m_n \ge 0\}}(M_{m_n}).$$

If these can be proved without assuming Li positivity or RH, then the program becomes a genuine proof strategy. Until then, it is best understood as a sharply formulated, falsifiable, and mathematically rich research program rather than a proof of RH.

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